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Case 7-2022: A 65-Year-Old Woman with Depression, Recurrent Falls, and Inability to Care for Herself

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PRESENTATION OF CASE

Dr. Diana L. Stern (Psychiatry): A 65-year-old woman was admitted to this hospital because of depression, recurrent falls, and difficulty caring for herself.

Four years before the current admission, the patient received a diagnosis of depression that was precipitated by psychosocial stressors. She was treated with bupropion, citalopram, and cognitive behavioral therapy. The depression was well controlled until 1 year before the current admission, when she began to have anhedonia, decreased energy, sadness, and poor concentration. She resumed cognitive behavioral therapy, but her symptoms worsened during the next 4 months, and she had suicidal ideation. The dose of bupropion was increased, and treatment with trazodone was started.

Fourteen weeks before the current admission, the patient was in a motor vehicle accident that was attributed to her falling asleep while driving. After the accident, she reported increased sadness, low energy, and inability to perform her usual activities. One week later, she fell and struck her head, and she was evaluated in the emergency department of this hospital. She reported that, before her fall, she had been feeling unsteady because of weakness in the left leg. A physical examination was normal. Computed tomography (CT) of the head, performed without the administration of intravenous contrast material, revealed hypodensities involving the subcortical and periventricular white matter, findings consistent with chronic small-vessel disease. The patient was referred to the neurology clinic of this hospital and was discharged home.

During the next week, the patient had balance difficulties and began using a walker. When she was evaluated in the neurology clinic, she reported imbalance and occasional nausea. On examination, she had a depressed affect. Strength was assessed as 5–/5 in the proximal and distal muscles of the arms and legs. Reflexes were 1+ and symmetric at the biceps, triceps, and patellar tendons and were

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absent at the ankles, with downgoing toes. The gait was wide-based and unsteady, with mild dragging of the left leg. A diagnosis of postconcussive syndrome was considered, and treatment with ondansetron was started for nausea. Additional tests were performed.

Magnetic resonance imaging (MRI) of the head, performed after the administration of intravenous contrast material, revealed increased signal intensity in the pons and in a periventricular distribution on T2-weighted imaging, without associated contrast enhancement. There was a region of decreased signal intensity in the right corona radiata on a map of the apparent diffusion coefficient. Results of electroencephalography were normal, as were results of electromyography and nerve-conduction studies of the left leg. Physical therapy was recommended, and repeat MRI was planned for 6 weeks later.

Over the next 6-week period, during sessions of cognitive behavioral therapy, the patient was noted to be less talkative, with slow processing, impaired recall, and mild inattention. Repeat MRI with angiography performed 6 weeks after the first study revealed patchy hyperintensity involving the centrum semiovale, corona radiata, internal capsule, and corticospinal tract bilaterally on T2-weighted fluid-attenuated inversion recovery (FLAIR) imaging, without contrast enhancement; the cerebral vasculature was normal.

The patient had recurrent falls without loss of consciousness and began using a wheelchair. She reported worsening sadness, anxiety, and anhedonia, and treatment with buspirone was started. On evaluation in the neurology clinic 6 weeks before the current admission, testing of leg strength yielded inconsistent results, but strength in both legs was assessed as 5-/5 with maximum effort. Dragging of the left leg was also noted to be inconsistent. There were concerns that her depression might be contributing to her neurologic changes, and further follow-up with the psychiatric service was recommended.

Five weeks before the current admission, the patient's husband brought her to the emergency department of this hospital for an evaluation of worsening memory impairment, poor self-care, and frequent falls. The patient had stopped participating in physical therapy and was spending most of the day in bed. She reported that she had lost 14 kg during the previous 4 months.

Other medical history included hypertension, diabetes, and osteoarthritis of the knees. There was a history of postpartum depression 27 years earlier that had lasted for several months. Medications included amlodipine, buspirone, bupropion, cholecalciferol, citalopram, metformin, and trazodone. There were no known drug allergies. The patient lived with her husband in a suburb of Boston and worked as a teacher. She was a lifelong nonsmoker, drank alcohol only occasionally, and did not use illicit drugs. Her mother had a history of anxiety and depression in her seventh decade of life. Her sister had a history of stroke.

On examination, the patient appeared tired and disheveled, with psychomotor retardation. Strength in both legs was assessed as 5/5. She had a monoplegic gait, with buckling of the left leg, when walking down a hallway, but she had a normal gait when walking on a treadmill. Her score on the Montreal Cognitive Assessment was 25, indicating mild cognitive impairment; scores range from 0 to 30, with higher scores indicating better cognitive function. The complete blood count, results of liver-function and kidney-function tests, and blood levels of cobalamin, thyrotropin, and electrolytes were normal. Screening for syphilis, Lyme disease, and the human immunodeficiency virus was negative, as was urine toxicologic screening. After an evaluation by a neurology consultant, the patient was voluntarily admitted to the inpatient psychiatric unit.

While the patient was in the hospital, the dose of citalopram was tapered, and treatment with desvenlafaxine and amphetamine-dextroamphetamine was started; her levels of attention and wakefulness increased. She participated in group psychotherapy, cognitive behavioral therapy, and occupational and physical therapy. She continued to have variable weakness in the left leg and had three witnessed falls, during which she reportedly had fatigue and lowered herself to the ground.

On hospital day 11, the patient reported being unable to use her left arm because of weakness. Although her left arm rested across her chest for most of the day, she was observed holding her husband's hand with her left hand when saying goodbye. On evaluation by a neurology consultant, strength in the left arm was assessed as 5/5. The working diagnosis was

functional neurologic disorder. On hospital day 12, the patient was discharged to a rehabilitation center.

At the rehabilitation center, the patient received physical, occupational, and speech therapy, as well as medications to treat depression. During the first week, she made progress toward independence in bathing, toileting, and grooming but had limited attention, memory, and problem solving. During the second week, she became more withdrawn. There was a paucity of speech with hypophonia, decreased attention, and less eye contact with visitors who sat on her left side than with visitors who sat on her right side. During the third week, the patient stopped answering questions and following commands. She was evaluated again in the emergency department of this hospital.

On examination, the temperature was 36.7°C, the blood pressure 141/90 mm Hg, the pulse 98 beats per minute, the respiratory rate 16 breaths per minute, and the oxygen saturation 99% while the patient was breathing ambient air. She was withdrawn and minimally interactive. She was able to state her name but not the date or location. She did not blink in response to threat on the left side and did not track the examiner's finger past the midline to the left. There was flattening of the left nasolabial fold. The left arm was hypertonic and held in flexion; there was no spontaneous movement of the left arm or leg, and she declined to participate in a strength examination. She did not withdraw the left arm in response to nail-bed pressure; she withdrew the left foot with antigravity movement in response to tactile stimulation.

The white-cell count was 12,940 per microliter (reference range, 4500 to 11,000), with a normal differential count. The remainder of the complete blood count was normal, as were the results of liver-function and kidney-function tests and blood levels of electrolytes.

A diagnostic test was performed.

INITIAL CLINICAL IMPRESSION

Dr. Judith A. Restrepo: When I evaluated this patient before her transfer to the rehabilitation center, I obtained a psychosocial history with collateral information and performed bedside neuropsychological screening, which formed the

basis of her psychiatric evaluation.¹ She had cognitive deficits in delayed recall and executive function, with insight into the changes. She also had changes in mobility, which can be a nidus for functional decline, resulting in hospitalization. Finally, she had changes in motivation, which may explain the inconsistencies noted in the examination.

Amotivation can result from abulia related to organic injury or disease, psychiatric disorders such as depression, or negative symptoms of schizophrenia or catatonia, or it can be volitional. This patient's fatigue with a flat affect in the context of cognitive deficits is suggestive of abulia. In the absence of a concrete organic cause identified on neurologic evaluation, functional syndromes — including functional neurologic disorder, chronic fatigue syndrome, and psychiatric phenomena — must be considered.² Depression was thought to be contributing to her presentation, but given the prominence of the physical changes, she was also treated for functional neurologic disorder.

Management of functional neurologic disorder includes providing the diagnosis as a form of therapy, treating coexisting psychiatric conditions, and using cognitive behavioral, physical, and occupational therapy to address cognitive and physical deficits.³ The patient's previous depressive episodes had resolved with routine medication and therapy, and the next step in the management of her depression involved transitioning to alternative antidepressant and augmentation agents. We elected to augment her antidepressant therapy with psychostimulants because of the fatigue and executive changes.^{4,5} Although the patient received appropriate treatment for functional neurologic disorder, her symptoms progressed while she was at the rehabilitation center, so her diagnosis was reassessed when she returned to the emergency department.

DIFFERENTIAL DIAGNOSIS

Dr. Deborah A. Forst: I was involved in the care of this patient, and I am aware of the final diagnosis in this case. This 65-year-old woman with a history of depression presented with worsening neuropsychiatric symptoms, weight loss, unsteady gait, and progression of weakness on the left side for several months. MRI of the head had

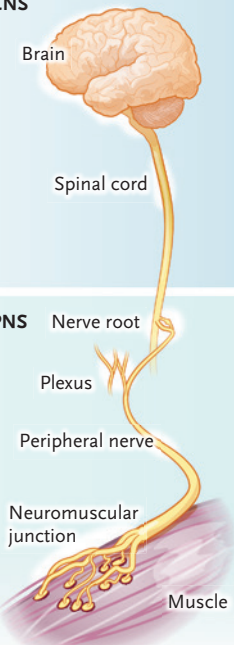
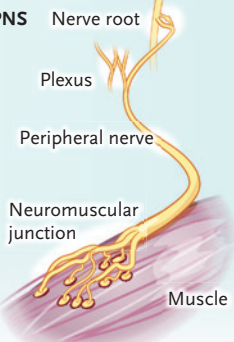
Neuraxis location	Patient's presenting signs and symptoms	Possible lesion localization or dysfunction
CNS 	Progressive spastic left hemiparesis involving face, arm, and leg	Localization to right parietal and frontal lobes (cortical and subcortical regions) CNS involvement on right side, above level of facial nucleus
	Left hemineglect	Right parietal lobe dysfunction
	Withdrawal, abulia, and anhedonia	Frontal lobe dysfunction
	Right-gaze preference	Inactivating lesion in right frontal eye field
	Possible left hemianopsia	Right retrochiasmal lesion
PNS 	Unilateral leg weakness with monoplegic gait	Primary PNS process The results of electromyography and nerve-conduction studies of the affected leg were unremarkable.
	Hyporeflexia in both legs	The history of diabetes mellitus provides an alternative explanation for the depressed reflexes, which may be unrelated to the current presentation.

Figure 1. Neuroanatomical Localization.

The initial step in evaluating the patient's neurologic illness was to localize the presenting signs and symptoms along the neuraxis. The features of her presentation and the findings on examination best corresponded to a disease process located in the central nervous system (CNS), involving the right frontal and parietal cortical and subcortical regions. Of note, the absence of blink in response to threat on the left side is consistent with either left hemineglect or left hemianopsia. PNS denotes peripheral nervous system.

revealed nonspecific findings. The patient received a diagnosis of functional neurologic disorder and was treated with inpatient psychiatric care, followed by rehabilitation. While she was at the rehabilitation center, she had progressive neurologic decline, which led to another evaluation in the emergency department of this hospital.

I will begin by considering the previous diagnosis of functional neurologic disorder. I will then discuss neuroanatomical localization and outline potential causes of this patient's illness.

FUNCTIONAL NEUROLOGIC DISORDER

This patient was initially thought to have functional neurologic disorder, a primary neuropsychiatric illness with neurologic manifestations. Her preexisting psychiatric disorder and the findings on physical examination supported this diagnosis. She was witnessed to have knee buckling with ambulation, which is commonly reported in patients with functional neurologic

disorder.⁶ She also had varying motor function in different settings; such motor inconsistency is reported to be 85.3 to 99.9% specific for functional neurologic disorder.⁶ However, because she had progressive neurologic decline at the rehabilitation center and markedly abnormal results on neurologic examination in the emergency department, her current presentation is more likely to be due to neuropsychiatric symptoms related to a primary progressive neurologic illness.

PRIMARY PROGRESSIVE NEUROLOGIC ILLNESS Neuroanatomical Localization

In evaluating any patient with a suspected primary neurologic illness, the first step is to localize the symptoms along the neuraxis (Fig. 1). This patient's initial symptoms, including unilateral leg weakness with a monoplegic gait and hyporeflexia in both legs, could be indicative of a process involving the peripheral nervous system (PNS). However, the results of electromyog-

raphy and nerve-conduction studies of the affected leg were unremarkable, findings that argue against a primary PNS process. Although the depressed reflexes in the legs are consistent with some degree of PNS dysfunction, the history of diabetes suggests an alternative cause of diabetic peripheral neuropathy, which may be unrelated to the current presentation.

Furthermore, the patient had signs and symptoms that were suggestive of a process involving the central nervous system (CNS). She had progressive spastic hemiparesis involving the face, arm, and leg on the left side, which localizes to the CNS on the right side, above the level of the facial nucleus. She had evidence of withdrawal, abulia, and anhedonia, features associated with frontal lobe involvement. She also had a right-gaze preference, which could suggest an inactivating lesion in the right frontal eye fields. There were findings suggestive of left hemineglect, which occurs in patients with right parietal dysfunction. She did not blink in response to threat on the left side; the absence of this response is consistent with either left hemineglect or left hemianopsia, the latter of which could be associated with a right retrochiasmal lesion. Ultimately, her neurologic signs and symptoms localize to the right frontal and parietal cortical and subcortical regions.

The differential diagnosis of diseases that may manifest with prominent neuropsychiatric symptoms is broad. In this patient, the list can be narrowed to diseases that manifest with subacute, progressive CNS decline with prominent neuropsychiatric manifestations, focal neurologic signs, and nonspecific MRI findings.

Neurodegenerative Diseases

Although genetic disorders that affect the white matter of the CNS classically develop in infancy or childhood, this patient could have an adult-onset leukodystrophy, such as Krabbe disease, vanishing white matter disease, or metachromatic leukodystrophy.^{7,8} Adult-onset Krabbe disease is an autosomal recessive lysosomal storage disease caused by β -galactocerebrosidase deficiency. Patients often present with spastic paraparesis or tetraparesis and with cognitive decline. MRI typically reveals hyperintensity involving the proximal corticospinal tract, optic radiation, and splenium or isthmus of the corpus callosum bilaterally on T2-weighted imaging, features that

could align with the nonspecific early MRI findings observed in this patient. However, the patient was older than would be expected for this diagnosis.⁷⁻⁹

Another possible explanation for this patient's presentation is CADASIL (cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy). CADASIL is the most common hereditary cause of cerebral small-vessel disease; it could be consistent with her sister's history of stroke. Focal neurologic signs, cognitive dysfunction, and neuropsychiatric symptoms are common presenting features of CADASIL. However, MRI did not reveal changes in the white matter at the temporal poles and external capsule, which are commonly seen in patients with this disease.¹⁰

Infection

Focal neurologic and neuropsychiatric symptoms, as well as nonspecific MRI findings, can occur in patients with indolent fungal infections such as cryptococcosis or aspergillosis¹¹; viral infections such as JC virus infection, which causes progressive multifocal leukoencephalopathy¹²; or prion diseases such as Creutzfeldt-Jakob disease.¹³ In this patient, MRI did not reveal findings typical of a fungal infection, such as abscess, space-occupying lesions, infarction, or hemorrhage from angioinvasion.^{11,14} She did not have impaired cellular immunity, a feature that would place her at risk for progressive multifocal leukoencephalopathy.¹² In addition, her course was more protracted than would be expected with Creutzfeldt-Jakob disease.¹³

Cerebral Venous Sinus Thrombosis

Cerebral venous sinus thrombosis is more common in women than in men, may be precipitated by head trauma, and can have a subacute course; 7 to 20% of patients present with symptoms that have lasted for more than 1 month.¹⁵⁻¹⁸ Although head trauma could have occurred in this patient during the motor vehicle accident, several aspects of her presentation make the diagnosis of cerebral venous sinus thrombosis unlikely. She did not report headache, which is present in 80 to 90% of patients.^{16,17} She is older than the typical patient (a woman of childbearing age).¹⁵ The slow trajectory of her symptoms would be highly unusual, given that the median time from symptom onset to the diagnosis of cerebral ve-

nous sinus thrombosis is 7 days.¹⁶ Finally, the MRI findings were not suggestive of venous infarction or thrombosis.

Autoimmune and Inflammatory Diseases

Autoimmune or inflammatory diseases of the CNS frequently have neurocognitive or neuropsychiatric manifestations. Disorders to consider in this patient include multiple sclerosis, primary angiitis of the CNS, and paraneoplastic encephalitis, although she did not have the classic presentation for any of these conditions.¹⁹⁻²² Neurosarcoidosis is an important consideration in this patient, given that it is associated with protean manifestations and varied imaging findings. Approximately 15% of patients with neurosarcoidosis present with parenchymal abnormalities, which were seen in this patient, and 60% of patients have depression.²³ This patient had no known history of systemic sarcoidosis, but only 31% of patients with neurosarcoidosis have concomitant systemic disease at the time of diagnosis.²³

Cancer

This patient's age, weight loss, and progressive course are suggestive of cancer. Patients with diffuse glioma or lymphoma involving the nervous system often present with insidious symptoms, and nonspecific findings such as neuropsychiatric symptoms may delay the diagnosis.²⁴⁻²⁶ However, the patchy bilateral hyperintensity observed on T2-weighted imaging in this patient is atypical of glioma.^{27,28} Of note, 40 to 50% of patients with primary CNS lymphoma present with neuropsychiatric, behavioral, or cognitive symptoms, and 50 to 70% of patients have focal neurologic deficits.²⁹⁻³¹ The median time from symptom onset to diagnosis is 3 to 5 months; in one series, 10% of diagnoses were established after a year or more.³¹ On the basis of the features of this patient's presentation, I thought that CNS lymphoma was the most likely diagnosis, although the absence of contrast enhancement on MRI would be rare.³² I recommended repeat MRI of the head with the administration of gadolinium as the next step in the evaluation of this patient.

DR. DEBORAH A. FORST'S DIAGNOSIS

Lymphoma of the central nervous system.

DIAGNOSTIC IMAGING

Dr. R. Gilberto Gonzalez: MRI of the head revealed an extensive, confluent, expansile abnormality that was associated with contrast enhancement and with hyperintensity on T2-weighted FLAIR imaging (Fig. 2). The abnormality originated from the centrum semiovale bilaterally, involved the right and left cingulate gyri, and extended across the body of the corpus callosum. There was an additional expansile abnormality, also associated with contrast enhancement and with hyperintensity on T2-weighted FLAIR imaging, within the posterior limb of the right internal capsule, extending into the right corticospinal tract. These enhancing lesions showed hyperintensity on diffusion-weighted imaging and hypointensity on a map of the apparent diffusion coefficient, features that indicate reduced diffusivity. There was additional confluent hyperintensity on T2-weighted FLAIR imaging, without contrast enhancement, involving the frontoparietal white matter, corpus callosum, basal ganglia and thalamus, internal capsule, pons, and midbrain bilaterally.

Dr. Forst: The MRI findings were highly suggestive of CNS lymphoma. I recommended a brain biopsy as the next step in diagnostic testing. The biopsy was to be performed before the administration of glucocorticoids, which can partially treat lymphoma and decrease the diagnostic yield of the biopsy.

DIAGNOSTIC PROCEDURE

Dr. Pamela S. Jones: Given the multifocal nature of the patient's lesions, we recommended a targeted tissue biopsy. We discussed with the patient and her family that the goal of surgery was to make a diagnosis and that the diagnostic yield of stereotactic brain biopsy is greater than 90%. We reviewed the risks of stereotactic biopsy, including postoperative complications (6 to 8%), permanent neurologic deficits (1 to 2%), and death (<1%).

We selected a right frontal approach because the area of contrast enhancement in the right frontal lobe was the largest target and came closest to the surface of the brain. We used frameless stereotactic navigation with the patient's MRI loaded into the image-guidance system (Fig. 3). We identified a target lesion site and an

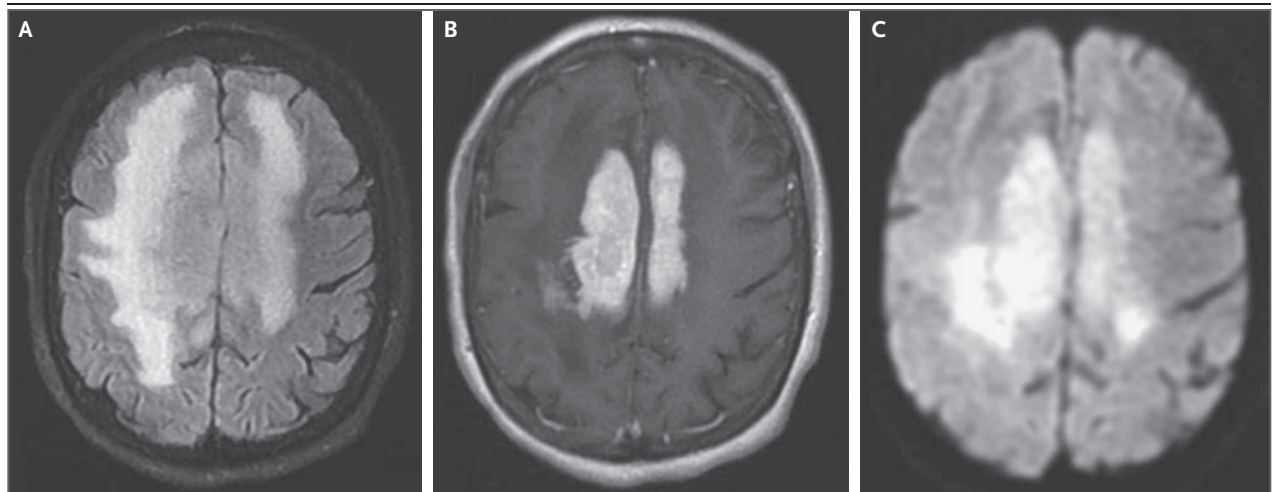


Figure 2. MRI of the Head.

When the patient returned to the emergency department after progressive neurologic decline occurred at the rehabilitation center, MRI of the head was performed. Multiple sequences were obtained at the level of the centrum semiovale. A T2-weighted fluid-attenuated inversion recovery (FLAIR) image (Panel A) shows an abnormality associated with two levels of increased signal intensity within the white matter, with higher signal intensity in the periphery. A T1-weighted image obtained after the administration of gadolinium (Panel B) shows intense contrast enhancement in the central portion of the abnormality, with surrounding nonenhancing hypointensity. A diffusion-weighted image (Panel C) shows reduced diffusivity in the central portion of the abnormality. This contrast-enhancing mass with surrounding edema has features characteristic of primary lymphoma of the central nervous system.

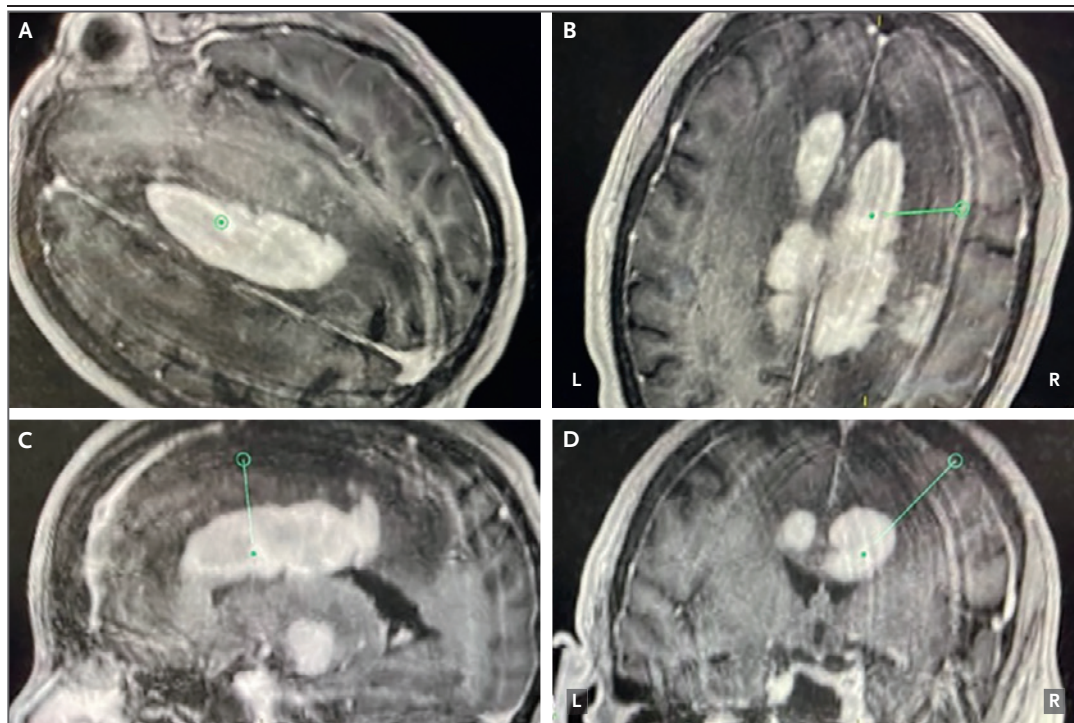


Figure 3. Images Guiding Targeted Tissue Biopsy of the Brain.

The patient's MRI was loaded into the image-guidance system, and a safe right frontal trajectory to the lesion was planned, as shown in green. The images show anatomical right and left, rather than radiologic right and left. Panel A shows the stopping point of the trajectory within the lesion. Panel B shows an axial view of the trajectory, Panel C a sagittal view, and Panel D a coronal view.

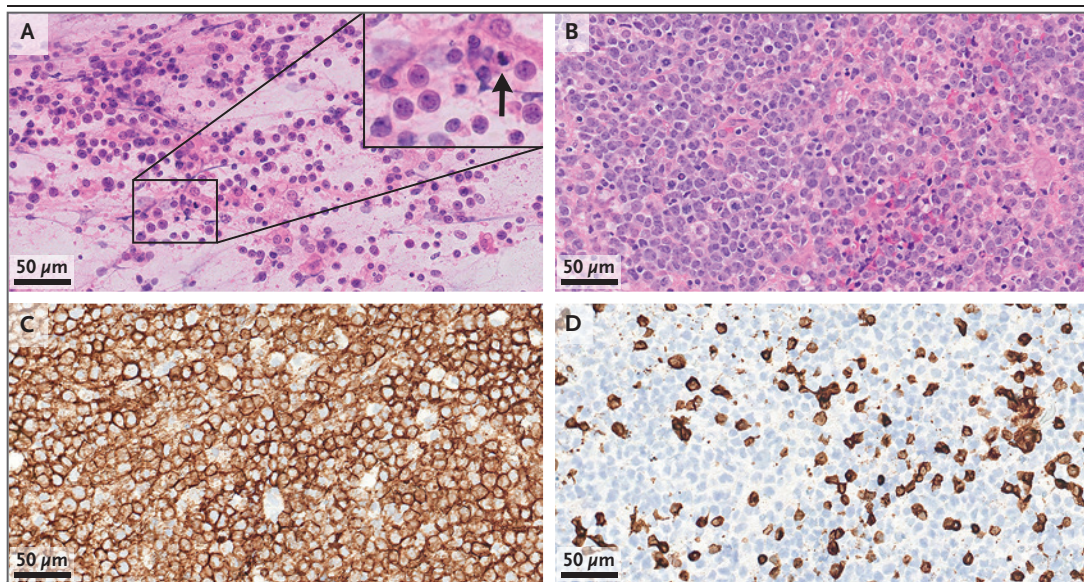


Figure 4. Biopsy Specimens of the Right Frontal Lobe.

On hematoxylin and eosin staining, a smear preparation for intraoperative consultation (Panel A) shows a hypercellular lesion composed of large, noncohesive cells with irregular nuclei, prominent nucleoli, and mitotic activity, features suggestive of a lymphoproliferative disorder; the inset shows the lesion at higher magnification, including a mitotic figure (arrow). Formalin-fixed, paraffin-embedded tissue (Panel B) shows a diffuse infiltrate of large atypical lymphoid cells with irregular-to-lobated nuclei, scant-to-moderate amounts of pale cytoplasm, and brisk mitotic activity. On immunohistochemical staining, the large atypical cells consist of CD20+ B cells (Panel C) admixed with scattered small CD3+ T cells (Panel D), features characteristic of a diffuse large B-cell lymphoma.

entry site, which was behind her hairline but anterior to the primary motor cortex.

We performed the needle biopsy through a small burr hole. Once we advanced the needle to the target lesion site, we obtained our first core biopsy sample and submitted it for frozen-section analysis. It was reviewed by our pathologists and determined to be lesional tissue. We then obtained two additional core biopsy samples and sent these for pathological evaluation. There were no complications during or after the procedure.

PATHOLOGICAL DISCUSSION

Dr. Michael S. Marshall: A small fragment of tan-white soft tissue was submitted for intraoperative consultation, and analyses of both a smear preparation and a frozen section were performed. The smear preparation showed a hypercellular lesion composed of large, noncohesive cells with irregular nuclei, prominent nucleoli, and mitotic activity (Fig. 4), features suggestive of a lymphoproliferative disorder. The frozen section showed similar large atypical lymphoid cells.

An examination of formalin-fixed, paraffin-embedded material revealed multiple fragments of tissue with a diffuse infiltrate of large atypical lymphoid cells with irregular-to-lobated nuclei, occasional prominent nucleoli, scant-to-moderate amounts of pale cytoplasm, and brisk mitotic activity. On immunohistochemical staining, the infiltrate consisted mostly of CD20+ B cells and scattered CD3+ T cells. Additional immunohistochemical stains showed that the large atypical cells coexpressed BCL6, BCL2, and MUM1 and were negative for CD10. Approximately 40% of the tumor cells were positive for C-MYC. In situ hybridization showed that the neoplastic cells were negative for Epstein–Barr virus–encoded RNA. A Ki-67 stain showed a proliferation index of greater than 90%. On the basis of diagnostic criteria with respect to immunohistochemical staining, these features were diagnostic of a diffuse large B-cell lymphoma, nongerminal center type.³³

PATHOLOGICAL DIAGNOSIS

Diffuse large B-cell lymphoma, nongerminal center type.

DISCUSSION OF MANAGEMENT
AND FOLLOW-UP

Dr. Forst: After receiving histopathological confirmation of CNS lymphoma, we performed additional testing to assess the extent of the patient's disease, which can inform treatment decisions and guide future monitoring for treatment response.^{29,34} The patient underwent positron-emission tomography and CT of the body to evaluate for systemic lymphoma. She also underwent a complete evaluation of the neuraxis, including a slit-lamp examination to evaluate for intraocular lymphoma, a lumbar puncture with cerebrospinal fluid analysis, and MRI of the spine. These examinations were all negative, confirming that the patient's disease was confined to the brain parenchyma.

The mainstay of treatment for primary CNS lymphoma includes induction chemotherapy with high-dose methotrexate, which is often combined with additional antineoplastic agents, given the evidence of higher response rates with the use of multidrug regimens.^{29,30} This patient received induction chemotherapy with high-dose methotrexate plus the anti-CD20 monoclonal antibody rituximab, an approach supported by a phase 2 study that showed a partial or complete radiographic response in 80% of patients treated with this regimen.³⁵ After four cycles of induction therapy, the patient had a complete response, defined as clearance of lymphoma from all affected compartments of the neuraxis without ongoing glucocorticoid use.³⁴ Because relapse

occurs in approximately half of patients with CNS lymphoma who have a response to induction therapy, typically within the first 2 years after treatment,^{29,36} consolidation strategies are used to decrease the risk of recurrence. In this patient, we were concerned about the neurocognitive side effects of whole-brain radiation therapy and about the intensity of myeloablative chemotherapy with autologous stem-cell rescue, so we opted for consolidation chemotherapy with etoposide and cytarabine, a regimen with proven efficacy against this disease.³⁷

The patient resided at an inpatient rehabilitation center for the duration of chemotherapy before ultimately returning home. She has had ongoing depression, anxiety, short-term memory loss, and inattention. She continues to receive psychiatric care and participates in cognitive behavioral therapy, cognitive-linguistic therapy, and physical therapy. She uses a single-point cane when walking, owing to moderate left hemiparesis. The most recent MRI of the head, performed 1 year after the completion of consolidation therapy, showed an ongoing complete response. She has been unable to return to work and receives long-term disability coverage.

FINAL DIAGNOSIS

Diffuse large B-cell lymphoma.

This case was presented at Neuroscience Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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